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(1) Research Program/Project/Study Title: Milkfish Belly Longganisa	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
Municipality/City:		ROXAS CITY							
Province:		CAPIZ							
Region:		VI							
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Food processing sector Restaurants Small and medium enterprise Academe		(8) Discipline/Field of Specialization Culinary							

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The use of meat substitutes has gained considerable attention in recent years, driven by concerns over environmental sustainability, health consciousness, and changing dietary preferences. According to Mihai Andrei (2022), one of the main reasons why Pork is bad is the composition of its high content of saturated fats and cholesterol that causes diabetes, cardiovascular disease, obesity, cirrhosis, and multiple types of cancer.

The study aimed to determine the Sensory Acceptability of Milkfish Belly Longganisa. Specifically, it aimed to. (1) determine the most preferred formulation among treatments, (2) General acceptability of milkfish belly longganisa among treatments, (3) Compare the overall sensory characteristics of the most with that of the control and (4) determine the acceptability level of the most preferred and the target consumers of Milkfish Belly longganisa as a meat substitute for longganisa among consumers. The study was conducted at Capiz State University, Dayao Satellite College.

Treatment 4 (100% Milkfish Belly) was evaluated as the most preferred formulation. In terms of general acceptability, it was evaluated as Extremely acceptable. In the most preferred attributes, Treatment 4 (100% Milkfish Belly) with the mean of 1.40 evaluated as the most acceptable treatment. The target populations of the study were adults and teens and evaluated the sample as "Like Moderately". There was no significant difference among treatments of Milkfish Belly Longganisa.

Further research is recommended to fully optimize the production of Milkfish Belly Longganisa. This includes conducting curing tests to assess product stability, additional physico-chemical analyses such as titration acidity determination, and moisture content analysis. Moreover, refining proper handling procedures during production and exploring potential substitutes for Milkfish Belly in the recipe are areas worthy of further investigation.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Frozen Milkfish Belly

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The concern for environmental sustainability and individual health has ignited a global shift towards plant-based and alternative protein sources. This dietary revolution has also extended its reach to the realm of traditional cuisines, prompting a reevaluation of cherished recipes. The Philippines, with its rich culinary heritage, is no exception. A prime example is longganisa, a beloved Filipino sausage, typically crafted from pork. While this delicacy holds a special place in the hearts of many, its production raises concerns about environmental impact and potential health implications linked to high levels of saturated fat and cholesterol.

A promising avenue to address these challenges lies in exploring alternative ingredients for longganisa production. One such candidate is the milkfish belly, a locally abundant and flavorful fish part. Given its availability and nutritional profile, it presents an enticing opportunity to create a sustainable and healthier version of this culinary staple without compromising taste or texture.

This study aims to investigate the feasibility of substituting milkfish belly for pork in longganisa production. By conducting sensory evaluations and nutritional analyses, the research will assess the acceptability of this novel product among consumers. Understanding the consumer response is crucial in determining the potential of milkfish belly longganisa as a viable alternative to the traditional pork-based variant.

Previous studies have explored the use of fish in sausage production, demonstrating its potential as a wholesome protein source. The Philippines' thriving milkfish industry, coupled with its historical integration into the local diet, provides a strong foundation for this research. While milkfish is rich in essential nutrients, it also contains fats, including some saturated fats and cholesterol. However, the presence of heart-healthy monounsaturated fats, particularly omega-3 fatty acids, offers a potential health advantage.

Furthermore, the growing emphasis on sustainable food production and the need to reduce waste in the food industry underscore the importance of exploring alternative uses for fish byproducts. By developing value-added products like milkfish longganisa, it is possible to create economic opportunities, reduce environmental impact, and meet the evolving demands of health-conscious consumers.

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Ultimately, this research seeks to contribute to the development of a more sustainable and nutritious food landscape while preserving the cultural significance of longganisa. By successfully introducing milkfish belly longganisa to the market, this study can potentially influence consumer preferences, inspire food producers to explore innovative alternatives, and inform policymakers about the potential benefits of promoting sustainable and healthier food choices.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials and Methods

Materials

The following are the list of materials used in this study are Knives, Basin, Weighing scale, Measuring spoon, Trays, Teaspoons, 4x10 plastic for packaging, Mixing bowl, Teaspoons, Chopping board, Cup and Measuring cups

Raw materials

The following are the raw materials used in the preparation of Milkfish Tocino: Milkfish belly, Water, pineapple juice, anisado wine, TVP, Garlic, black pepper, phosphate, curing Salt, MSG and Salt.

Experimental Treatment

Treatment	Control	T1	T2	T3	T4
Pork	100%	75%	50%	25%	
Milkfish belly		25%	50%	75%	100%

* Other Ingredients is constant

METHODS

Preparation of Milkfish belly

The milkfish deboning process started with washing the freshly harvested milkfish. It was then split (butterfly cut), and internal organs, including the bile, were removed. Once this was done, the deboning process followed suit. In the deboning procedure, the spines (dorsal, ventral, lateral, and intramuscular spines) and rib bones were removed gradually using sterile mosquito forceps. To improve the taste, deboned milkfish was then marinated with assorted seasonings before it was vacuum-packed in a polyethylene bag and stored in the freezer.

Preparation of Milkfish belly longganisa

1. Weigh 1000 grams of Milkfish belly and set it aside.
2. Mix salt and cure salt in the milkfish belly.
3. Add phosphate dissolved in water and mix very well.
4. Mix the other dry ingredients well and add them to the Milkfish belly.
5. Add the liquid ingredients.
6. Mix well until uniformly blended.
7. Cure in the refrigerator for 3–4 days, turning every day to a low, uniform curing.
8. Pack in PP bags and store in the freezer until needed.

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Experimental Design and Treatment

Experimental Design

This study used a single-factor experiment in a completely randomized design (CRD) with four (4) treatments, and each treatment had three (2) replications. The Analysis of Variance (ANOVA) was used to analyze the data.

Experimental Treatment

The treatments were as follows: The proportion of the longganisa was divided into the following ratios per 1000g: Formulation 1, which is 75% ground pork + 25% Milkfish belly; Formulation 2, which is composed of 50% ground pork + 50% Milkfish belly; Formulation 3, which is 25% ground pork + 75% Milkfish belly; and Formulation 4, which is 100% milkfish belly, which is also our control group.

Data Gathering Procedure

The data was gathered using score cards, where ratings were indicated in terms of general acceptability of the product, most preferred formulation, and a preference test. The evaluators in this study were ten (10) semi-trained panelists from Food Technology students and teachers at CapSU Dayao Satellite College, and One Hundred (100) panelists were composed of children (5–12 years old), teens (13–19 years old), adults (20–39 years old), and middle-aged adults (40–59 years old).

Data Analysis Procedures

The Statistical Package for Social Sciences (SPSS) generated and processed all the data needed in this study. The data was analyzed using the mean and analysis of variance (ANOVA). The weighted mean was used to determine the consumer's acceptability of the Milkfish belly longganisa.

Statistical Tools and Analysis

The data were collected and collated in a tabulation sheet for analysis and interpretation. The statistical tools used in the analysis of data were mean and ANOVA. Mean was computed to determine the significant differences in the sensory qualities of Milkfish tocino among the four (4) treatments.

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(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussion

Results and Discussion

Table 1 Mean Rank of Most preferred formulation (MPF) Milkfish Tocino

Trial	Treatment A	Treatment B	Treatment C	Treatment D
1	2.6	3.2	1.8	2.7
2	2.5	3	2.2	2.2
Mean	2.55	3.1	2	2.45

Based on the mean rank, treatment C has the highest rank, which makes it the most preferred formulation with 25% pork and 75% milkfish belly.

Table 2a Friedman's Test Results of the four formulations

N	20
Chi-square	7.571
Df.	3
Asymp. Sig.	0.56

With a 5% level of confidence the result of Friedman test shows that there was no significant difference of the rank of 4 treatments. Therefore, you can use the four (4) treatments with 250 grams debone milkfish, 750 grams milkfish meat and 1000 grams debone milkfish except in Treatment B with 500 grams of debone milkfish in making the Milkfish Tocino.

Table 2b. General Acceptability of Different Treatment Milkfish Tocino with Natural Color

Treatment	Replication 1	Replication 2	Mean	Verbal Interpretation
1	7.7	7.7	7.7	LM (Like Moderately)
2	7.2	7.3	7.25	LM (Like Moderately)
3	8.2	8	8.1	LVM (Like Very Much)
4	7.9	7.6	7.75	LM (Like Moderately)

The total of (4) four different formulations are shown in table 4 In Treatment A, Formulation 1 (250 grams of debone milkfish) has 7.7, Treatment B, Formulation 2 (500 grams of debone milkfish) has a total of 7.25, Treatment C, Formulation 3 (750 grams of debone milkfish) has a total mean of 8.1, Treatment D Formulation 4 (1000 grams of debone milkfish) has a total mean of 7.75 .Therefore, treatment C got the highest acceptability level of 8.1 (750 grams of debone milkfish) with verbal interpretation of "like very much".

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Table 3 Sensory characteristics of most acceptable treatment in terms of color, odor, texture and taste

	Trial 1	Trial 2	Mean	Verbal Interpreting
Color	7.05	11.32	9.18	Naïve Peach
Odor	12.06	14.82	13.71	Extremely Fishy Odor
Texture	11.58	10.30	10.94	Extremely Soft and moist
Taste	12.75	10.4	11.57	Moderately sweet

Table 3 shows that the Sensory Acceptability of Milkfish Tocino in terms of Color, Treatment C (750g debone milkfish) on the first trial had a score of 7.05 and got 11.32 on the second trial. Overall, the color of the most preferred formulation of Debone Milkfish Tocino with a mean of 9.18 which is described as “Naïve peach”.

In terms of odor, Treatment C (750g debone milkfish) on the first trial had a score of 12.06 and got 14.82 on the second trial. Overall, the odor of the most preferred formulation of Debone Milkfish Tocino with a mean of 13.71 which is described as “Extremely Fishy Odor”.

In terms of Texture, Treatment C (750g debone milkfish) on the first trial had a score of 11.58 and got 10.30 on the second trial. Overall, the texture of the most preferred formulation of debone milkfish tocino had a mean of 10. 94 which is described as “extremely soft and moist”.

In Terms of Taste, Treatment C (750g debone milkfish) on the first trial had a score of 12.75 and got 10.4 on the second trial. Overall, the taste of the most preferred formulation of debone milkfish tocino had a mean of 11.57 which is described as “Moderately Sweet”.

Therefore, the sensory characteristics of Milkfish Tocino in terms of color was “Naïve peach”, in terms of odor was “Extremely Fishy Odor”, in terms of texture was “extremely soft and moist”, and in terms of taste was “moderately sweet”.

Table 4 Consumer acceptability of the most preferred treatment

Consumers' Category	Mean	Verbal Interpretation
Child (5-12 years old)	8.76	Like Extremely
Teen (13-19 years old)	8.4	Like Extremely
Adult (20-39 years old)	8.2	Like Extremely
Middle Age Adult (40- 59 years old)	8.32	Like Extremely
Mean	8.42	Like Extremely

Results in table 4 shows the consumer test for general acceptability of the Most Preferred Formulation (MPF) Milkfish Tocino shown in Table 7. Children (5-12 years old) gained rating 8.76 like extremely, Teens (13-19 years old) gained rating 8.40 like extremely, Adult (20-39 years old) gained rating 8.20 like extremely, Middle Age Adult (40-59 years old) gained rating 8.32 like extremely. Therefore, there was no significant difference between four (4) different groups of respondents. (See Appendix B.5) Overall according to the data gathered, child, teens and adults were the study's target population since they gave the sample a "Like Extremely" rating.

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSION AND RECOMMENDATIONS

The most preferred formulation (MPF) among treatments of Milkfish Tocino was Treatment C (750 grams debone milkfish). The sensory acceptability of the most preferred treatment for Milkfish Tocino was rated as "Naïve Peach" in terms of color, "Extremely fishy odor" in terms of odor, "Extremely soft and moist" in terms of texture, and "Moderately Sweet" in terms of taste.

The acceptability rating of the MPF of the Milkfish Tocino is between neither like nor dislike to like extremely among the four (4) age groups, the results shows on the data we gathered that the target consumers were child, teens and adults with a rating of "Like very much".

It is also recommended to use qualitative data, such as open-ended questions or focus group discussions, to gather more detailed feedback on the taste, texture, aroma, and appearance of Treatment C. And it would be beneficial to compare Treatment C with other existing products or variations to determine its competitive advantage and potential market positioning.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The study aimed to conduct product development on Bangus belly Longanisa. It was conducted at Capiz State University Fisheries and Food Research Development Center, Dayao Satellite College last May, 2024. The study used 4 treatments (4) with three (3) replications. Ten (10) panel of experts evaluated the sensory qualities of the products, and One hundred (100) panelist for consumer test (50 college student and 50 employee).

(15) Technology Transfer Activity (Indicate the specific products, activity design, processes or services transferred as extension activities)

Production of Milkfish Belly Longganisa. Process will be taught to the target clientele in the community through extension program of the college. Present the technology to the community and entrepreneur. Present the study in research symposium. Milkfish Belly Longganisa for family consumption and backyard micro industry.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

My deepest appreciation goes to the Capiz State University, FFRDC their facilities, technical support, and research environment were instrumental in the conduct of this study. Special thanks to Dr. Editha C. Alfon and Dr. Sebastian C. Caduco for funding this research

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I am grateful also to Dr. Janet Ledesma for her constructive feedback and suggestions, which greatly improved the clarity and impact of our study.

Lastly, I would like to thank the participants who took part in the sensory evaluation. Their honest feedback and enthusiastic participation were essential to the success of this research.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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